

Supplementary experiments of rebuttal

A High-dimensional Imbalanced Datasets

The supplementary experiments are conducted on two high-dimensional imbalanced datasets: PC2, which contains 745 samples with 36 features and an imbalance ratio of 45.562, and MC1, which comprises 1,988 samples with 38 features and an imbalance ratio of 42.217.

Table 1: Performance comparison of different oversampling methods on RF

Metric	Dataset	stableSMOTE	TabDDPM	CoDi	TabDiT	RelDDPM	GOIO	TabMaDo(Ours)
F1	MC1	0.4754 ± 0.023	0.5095 ± 0.041	0.5329 ± 0.073	0.6013 ± 0.007	0.4693 ± 0.021	0.6197 ± 0.000	0.6199 ± 0.000
	PC2	0.7995 ± 0.016	0.8103 ± 0.018	0.7748 ± 0.017	0.8603 ± 0.009	0.8010 ± 0.015	0.8363 ± 0.011	0.8905 ± 0.009
G-mean	MC1	0.8204 ± 0.062	0.8426 ± 0.075	0.8580 ± 0.071	0.8883 ± 0.066	0.8110 ± 0.080	0.8862 ± 0.086	0.8999 ± 0.044
	PC2	0.9098 ± 0.023	0.9226 ± 0.021	0.8982 ± 0.015	0.9208 ± 0.007	0.9070 ± 0.031	0.9265 ± 0.014	0.9336 ± 0.011
AUC	MC1	0.7490 ± 0.032	0.7803 ± 0.048	0.7082 ± 0.040	0.7142 ± 0.038	0.7515 ± 0.050	0.8143 ± 0.050	0.7303 ± 0.042
	PC2	0.8488 ± 0.021	0.8569 ± 0.015	0.8280 ± 0.031	0.8922 ± 0.010	0.8489 ± 0.019	0.8720 ± 0.007	0.9163 ± 0.004

Table 2: Performance comparison of different oversampling methods on GBDT

Metric	Dataset	stableSMOTE	TabDDPM	CoDi	TabDiT	RelDDPM	GOIO	TabMaDo(Ours)
F1	MC1	0.6451 ± 0.015	0.5282 ± 0.030	0.6601 ± 0.012	0.6667 ± 0.031	0.5746 ± 0.024	0.6147 ± 0.029	0.6675 ± 0.024
	PC2	0.7549 ± 0.066	0.7501 ± 0.075	0.7207 ± 0.045	0.7655 ± 0.064	0.7609 ± 0.053	0.7292 ± 0.070	0.7686 ± 0.046
G-mean	MC1	0.8839 ± 0.018	0.8211 ± 0.028	0.8736 ± 0.018	0.8833 ± 0.018	0.8300 ± 0.048	0.8834 ± 0.026	0.8906 ± 0.017
	PC2	0.9142 ± 0.032	0.8922 ± 0.065	0.8969 ± 0.047	0.9173 ± 0.061	0.9079 ± 0.065	0.9013 ± 0.050	0.9227 ± 0.085
AUC	MC1	0.8404 ± 0.022	0.7297 ± 0.054	0.8427 ± 0.081	0.8443 ± 0.038	0.7769 ± 0.035	0.8322 ± 0.096	0.8479 ± 0.025
	PC2	0.8088 ± 0.014	0.8041 ± 0.067	0.7762 ± 0.055	0.8141 ± 0.285	0.8152 ± 0.031	0.7841 ± 0.188	0.8157 ± 0.028

Table 3: Performance comparison of different oversampling methods on TabPFN

Metric	Dataset	stableSMOTE	TabDDPM	CoDi	TabDiT	RelDDPM	GOIO	TabMaDo(Ours)
F1	MC1	0.6537 ± 0.0511	0.6641 ± 0.0937	0.5767 ± 0.0305	0.6919 ± 0.1414	0.5254 ± 0.1041	0.5387 ± 0.1944	0.6981 ± 0.1515
	PC2	0.7243 ± 0.0337	0.7740 ± 0.0440	0.7477 ± 0.0095	0.7831 ± 0.1267	0.7596 ± 0.0332	0.7875 ± 0.0714	0.8234 ± 0.0975
G-mean	MC1	0.8555 ± 0.0500	0.8024 ± 0.1229	0.7447 ± 0.0727	0.8690 ± 0.0783	0.7931 ± 0.1066	0.7978 ± 0.1321	0.8706 ± 0.0735
	PC2	0.7867 ± 0.0201	0.8269 ± 0.0503	0.8083 ± 0.1532	0.8338 ± 0.3332	0.8175 ± 0.0380	0.8398 ± 0.2053	0.8634 ± 0.1578
AUC	MC1	0.7403 ± 0.0797	0.7063 ± 0.1053	0.6118 ± 0.1234	0.8086 ± 0.0936	0.7101 ± 0.1200	0.7166 ± 0.0786	0.8111 ± 0.0940
	PC2	0.7802 ± 0.0790	0.8263 ± 0.0779	0.8043 ± 0.0626	0.8316 ± 0.0672	0.8121 ± 0.0968	0.8346 ± 0.0639	0.8601 ± 0.0858

B Latest Tabular Generation Method: GBOT

We additionally include GBOT as a competitive baseline. GBOT employs a diffusion GAN-based framework to generate synthetic samples for imbalanced tabular datasets, demonstrating competitive performance in previous studies. The experimental results are reported with the same evaluation protocol as other methods to ensure fair comparison.

Table 4: Performance of RF

Dataset	F1-score	G-mean	AUC
yeast6	0.472 ± 0.023	0.698 ± 0.040	0.891 ± 0.052
flare-F	0.250 ± 0.048	0.493 ± 0.100	0.824 ± 0.202
abalone9-18	0.207 ± 0.059	0.646 ± 0.063	0.777 ± 0.054
glass-0-1-4-6-vs-2	0.200 ± 0.084	0.562 ± 0.203	0.540 ± 0.226
vehicle0	0.458 ± 0.068	0.637 ± 0.070	0.740 ± 0.073
german	0.333 ± 0.113	0.478 ± 0.136	0.683 ± 0.107
saheart	0.571 ± 0.113	0.666 ± 0.094	0.753 ± 0.113
pima	0.607 ± 0.194	0.692 ± 0.103	0.747 ± 0.191
magic	0.455 ± 0.014	0.568 ± 0.012	0.690 ± 0.010
MC1	0.601 ± 0.073	0.819 ± 0.082	0.734 ± 0.040
PC2	0.821 ± 0.014	0.937 ± 0.053	0.873 ± 0.024

Table 5: Performance of GBDT

Dataset	F1-score	G-mean	AUC
yeast6	0.536 ± 0.121	0.684 ± 0.040	0.876 ± 0.194
flare-F	0.182 ± 0.147	0.485 ± 0.093	0.727 ± 0.119
abalone9-18	0.500 ± 0.054	0.628 ± 0.020	0.809 ± 0.097
glass-0-1-4-6-vs-2	0.500 ± 0.298	0.688 ± 0.089	0.658 ± 0.242
vehicle0	0.950 ± 0.053	0.967 ± 0.090	0.997 ± 0.071
german	0.326 ± 0.118	0.462 ± 0.125	0.730 ± 0.129
saheart	0.435 ± 0.091	0.541 ± 0.078	0.707 ± 0.122
pima	0.556 ± 0.187	0.650 ± 0.112	0.793 ± 0.230
magic	0.805 ± 0.012	0.837 ± 0.012	0.921 ± 0.016
MC1	0.651 ± 0.245	0.880 ± 0.062	0.790 ± 0.133
PC2	0.756 ± 0.027	0.927 ± 0.024	0.813 ± 0.020

Table 6: Performance of TabPFN

Dataset	F1-score	G-mean	AUC
yeast6	0.400 ± 0.143	0.500 ± 0.068	0.891 ± 0.071
flare-F	0.343 ± 0.150	0.649 ± 0.034	0.879 ± 0.038
abalone9-18	0.200 ± 0.082	0.434 ± 0.087	0.794 ± 0.084
glass-0-1-4-6-vs-2	0.333 ± 0.305	0.649 ± 0.068	0.763 ± 0.076
vehicle0	0.927 ± 0.073	0.960 ± 0.014	0.994 ± 0.022
german	0.597 ± 0.091	0.697 ± 0.104	0.834 ± 0.072
saheart	0.444 ± 0.101	0.561 ± 0.035	0.732 ± 0.015
pima	0.679 ± 0.181	0.750 ± 0.067	0.829 ± 0.067
magic	0.823 ± 0.010	0.853 ± 0.036	0.930 ± 0.044
MC1	0.662 ± 0.100	0.885 ± 0.007	0.792 ± 0.049
PC2	0.819 ± 0.018	0.840 ± 0.098	0.860 ± 0.094

C Training Loss Curves of Guider

As depicted in Figure 1, the training process of the Guider demonstrates stable convergence behavior. The classification loss L_{class} and prototype loss L_{proto} decrease steadily throughout the training, with minor fluctuations that gradually diminish. The total loss $L_G = L_{class} + \gamma L_{proto}$ follows a similar trend, confirming that the proposed objective function effectively guides the model toward an optimal solution without overfitting.

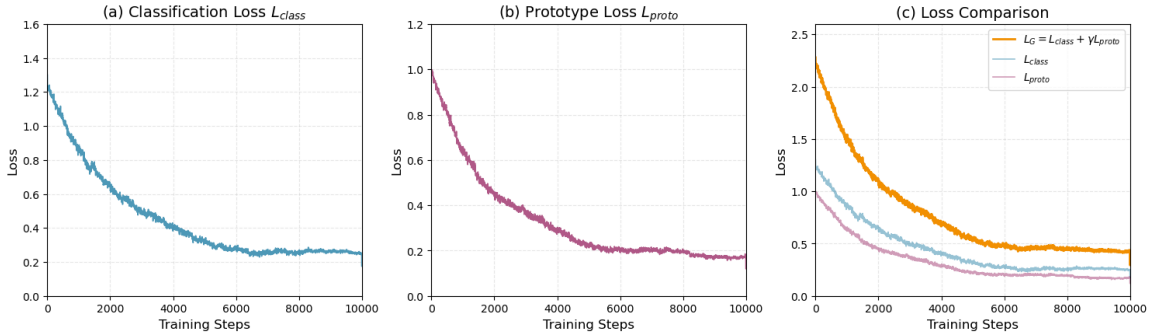


Figure 1: Convergence curves of the loss functions for the Guider. (a) Classification loss L_{class} . (b) Prototype loss L_{proto} . (c) Comparison of loss components, where the total loss is defined as $L_G = L_{class} + \gamma L_{proto}$.

D Ablation Study of Transformer

To investigate the contribution of the TabuLamba component in our proposed framework, we conduct an ablation study by replacing TabuLamba with a standard Transformer encoder of comparable parameter size. This variant is denoted as V5. All other components and hyperparameters remain identical to our full model to ensure a fair comparison.

Table 7: Ablation study results: comparison between our full model and V5

Dataset	F1-score	G-mean	AUC
yeast6	0.4330 ± 0.069	0.6631 ± 0.135	0.8530 ± 0.138
Wilt	0.8125 ± 0.087	0.9027 ± 0.091	0.9733 ± 0.111
abalone9-18	0.3820 ± 0.191	0.6176 ± 0.110	0.7463 ± 0.220
ecoli4	0.7410 ± 0.008	0.7824 ± 0.009	0.9729 ± 0.013
glass-0-1-4-6-vs-2	0.3361 ± 0.083	0.5211 ± 0.053	0.8177 ± 0.083
vehicle0	0.9165 ± 0.012	0.9402 ± 0.018	0.9818 ± 0.017
haberman	0.4181 ± 0.094	0.6062 ± 0.052	0.6351 ± 0.109
phoneme	0.7894 ± 0.026	0.8413 ± 0.041	0.9470 ± 0.043
german	0.5130 ± 0.046	0.6585 ± 0.070	0.7386 ± 0.076
saheart	0.4792 ± 0.044	0.5813 ± 0.031	0.6903 ± 0.020
magic	0.7891 ± 0.042	0.8264 ± 0.038	0.9127 ± 0.201